CSIS 1280 · MODULES 1 TO 6

Trailhead Code Lab

Student Guide: how the lab works and what you are expected to do

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Ten hands-on HTML and CSS activities · about 2½ to 3 hours in total

Open the lab at **https://rezaghaeli.github.io/csis-1280-code-lab/**

Read this guide once before you start. It takes about ten minutes.

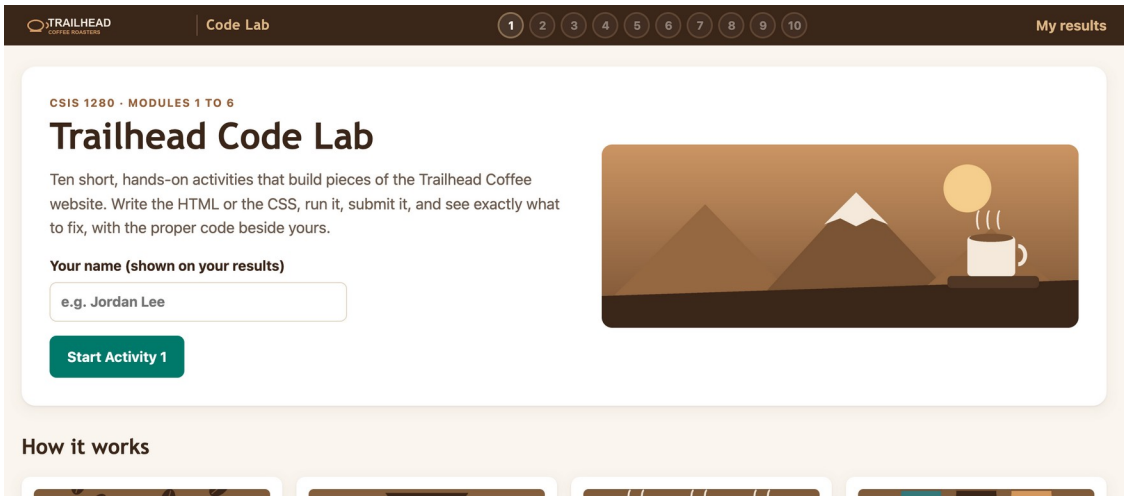


Figure 1. The Trailhead Code Lab home page.

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1. What Is the Trailhead Code Lab?

The Trailhead Code Lab is a website where you practise everything from Modules 1 to 6 by **writing real code**. It continues the Trailhead Coffee Roasters story from the case studies: each activity asks you to build one piece of the café's website, from the HTML skeleton of the home page to a booking form.

In every activity you:

- read a short task with numbered steps;
- write either the **HTML** or the **CSS** (the other file is already written for you);
- watch your page change as you type, and compare it with a picture of the goal;
- **submit** your code and get an instant report: a percentage score, each mistake explained, and the proper code beside yours.

Submitting an activity unlocks the next one. When you finish all ten, you get an overall score for the full cycle. You can do the whole lab again as many times as you like.

The ten activities

#	Module	You write	What you build	About
1	1	HTML	The home page skeleton: header, nav, sections, image, footer	15 min
2	1	HTML	A pour-over brew guide: lists, a nested list, a link, a quote	15 min
3	2	CSS	Menu colours and fonts with five kinds of selector	15 min
4	2	CSS	A "Daily Special" card: padding, border, margin, box-sizing, em	15 min
5	3	CSS	Text wrapped around a floated photo, and clear	10 min
6	4	CSS	The café menu as a CSS Grid	15 min
7	5	CSS	A header and class cards with Flexbox	15 min
8	5	CSS	A phone layout with a media query	15 min
9	6	HTML	The roastery tour video with captions, and the podcast	15 min
10	6	HTML	The Book a Class form, with validation	20 min

> **Why do it?**
>
> Reading about CSS is not the same as writing it. The lab gives you ten small, safe places to make mistakes and see straight away **why** something did not work.
>
> Each check matches something you will meet again in labs, assignments, and quizzes.

2. What You Are Expected to Do

Unless your instructor tells you otherwise, this is what is expected:

- ☐ Complete **all ten activities, in order**. Each one unlocks when you submit the one before it.
- ☐ Write the code **yourself**. Use the steps, your class notes, and the case studies. Look at the proper code only **after** you submit; that is when it helps you learn.
- ☐ **Read every report.** For each mistake, read *We found*, *Why*, and *The correct code*, then fix it in your own code.
- ☐ If an activity scores **below 80%**, fix your code and **submit it again** before moving on (or come back to it later).
- ☐ Aim for an **overall score of 80% or more** for the full cycle.
- ☐ When you finish, **save your results page as a PDF** (Section 9) and upload it to the course page if your instructor asks for it.

> **How your score is worked out**
>
> Each activity has 6 to 9 checks, and each check is worth 1 to 3 points. Your activity score is the points you earned as a percentage. Your **overall** score is the average of the ten activity scores.
>
> The score is **approximate**: the lab checks the important parts of each task, not every detail of your code. A page that looks right but breaks a rule (for example, an image with no `alt` text) still loses points.
>
> You can submit an activity as many times as you like. Your **latest** submission counts.

3. Before You Start

Open the lab

Open this address in your browser, or scan the code with your phone to save the link:

https://rezaghaeli.github.io/csis-1280-code-lab/



Figure 2. Scan to open the Trailhead Code Lab.

What you need

- A **laptop or desktop computer** with an up-to-date Chrome, Edge, Firefox, or Safari. You can read the activities on a phone, but typing code on one is hard.
- Nothing to install, no account, and no password.

> **Your progress is saved in your browser**
>
> The lab remembers your code and scores **in the browser you use**: there is no account. So:
>
> - Use the **same computer and the same browser** each time.
> - Do **not** use a private or incognito window: everything is forgotten when it closes.
> - Clearing your browser history or “site data” also clears your lab progress.
>
> Nothing you type is sent anywhere. Your instructor sees your results only if you send them the PDF.

4. A Tour of the Home Page

The home page is where you start, and where you come back to see your progress.

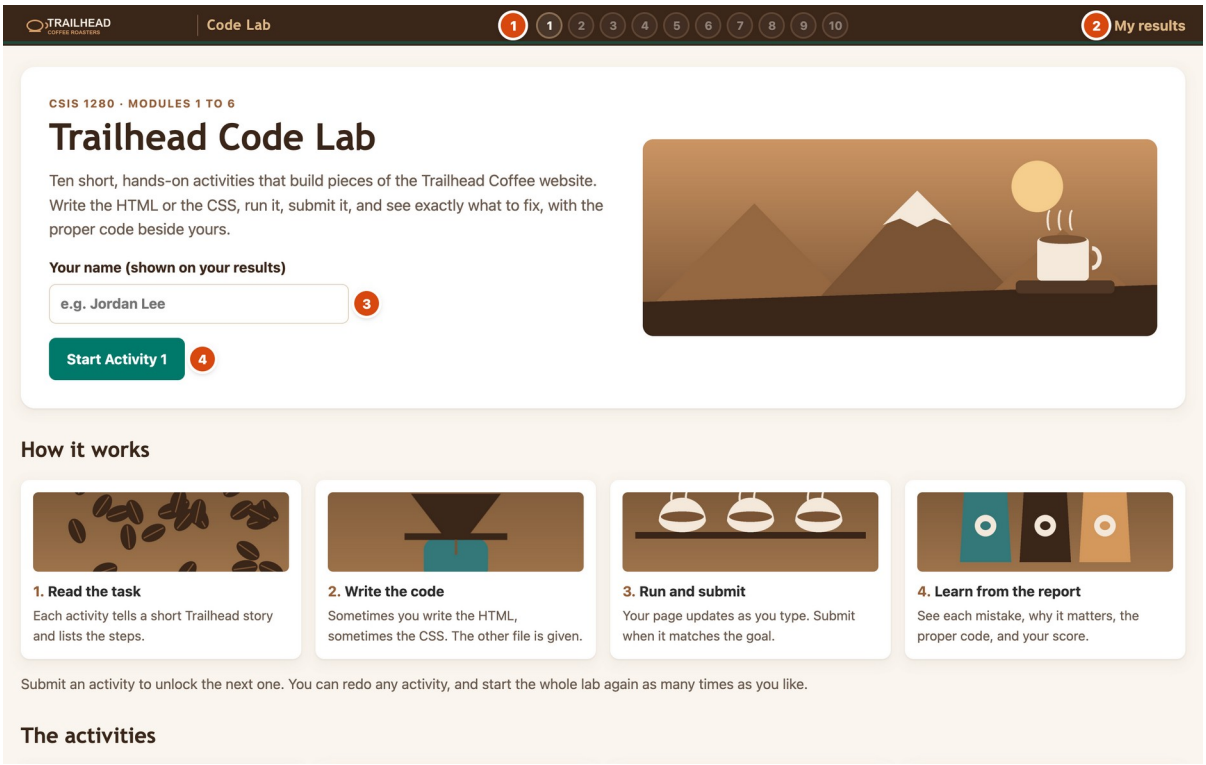


Figure 3. The home page. The numbers match the table below.

#	What it is	What to do with it
1	The activity circles	One circle per activity. Click an open one to go to it. They change colour as you finish activities (Section 8).
2	My results	Your scores so far, and the full-cycle score when you finish.
3	Your name	Type your name. It appears on your results page and PDF.
4	Start / Continue	Takes you to the next activity you have not submitted yet.

Scroll down to see **How it works** and the list of activities. Only the next activity is **Open**; the rest show a padlock until you get there.

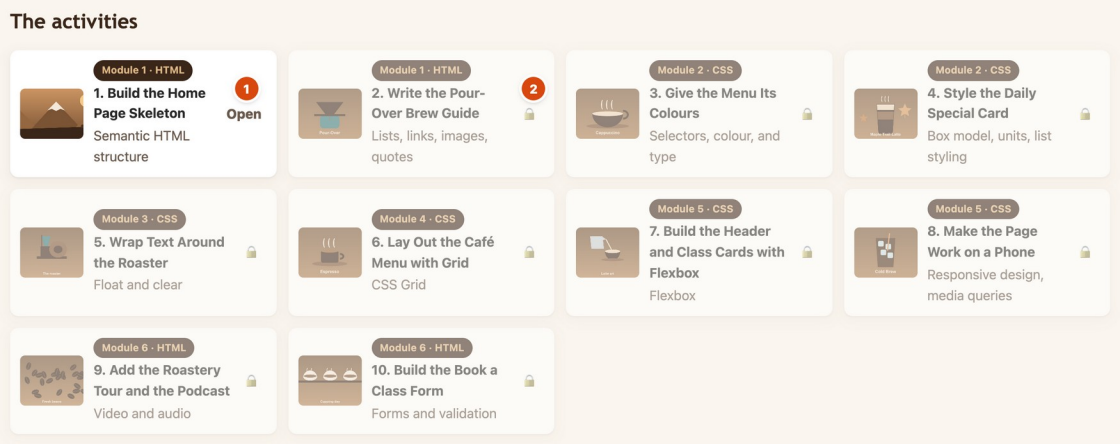


Figure 4. Activity 1 is open (1); the others are locked (2) until you submit the one before.

If you try to open a locked activity, the lab tells you which one to submit first:

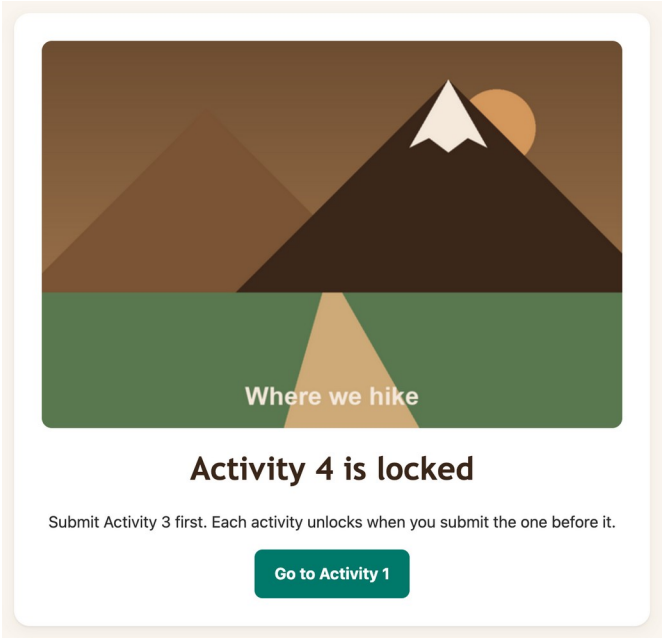


Figure 5. A locked activity.

5. Inside an Activity

Every activity uses the same screen: the task on the left and your workspace on the right.

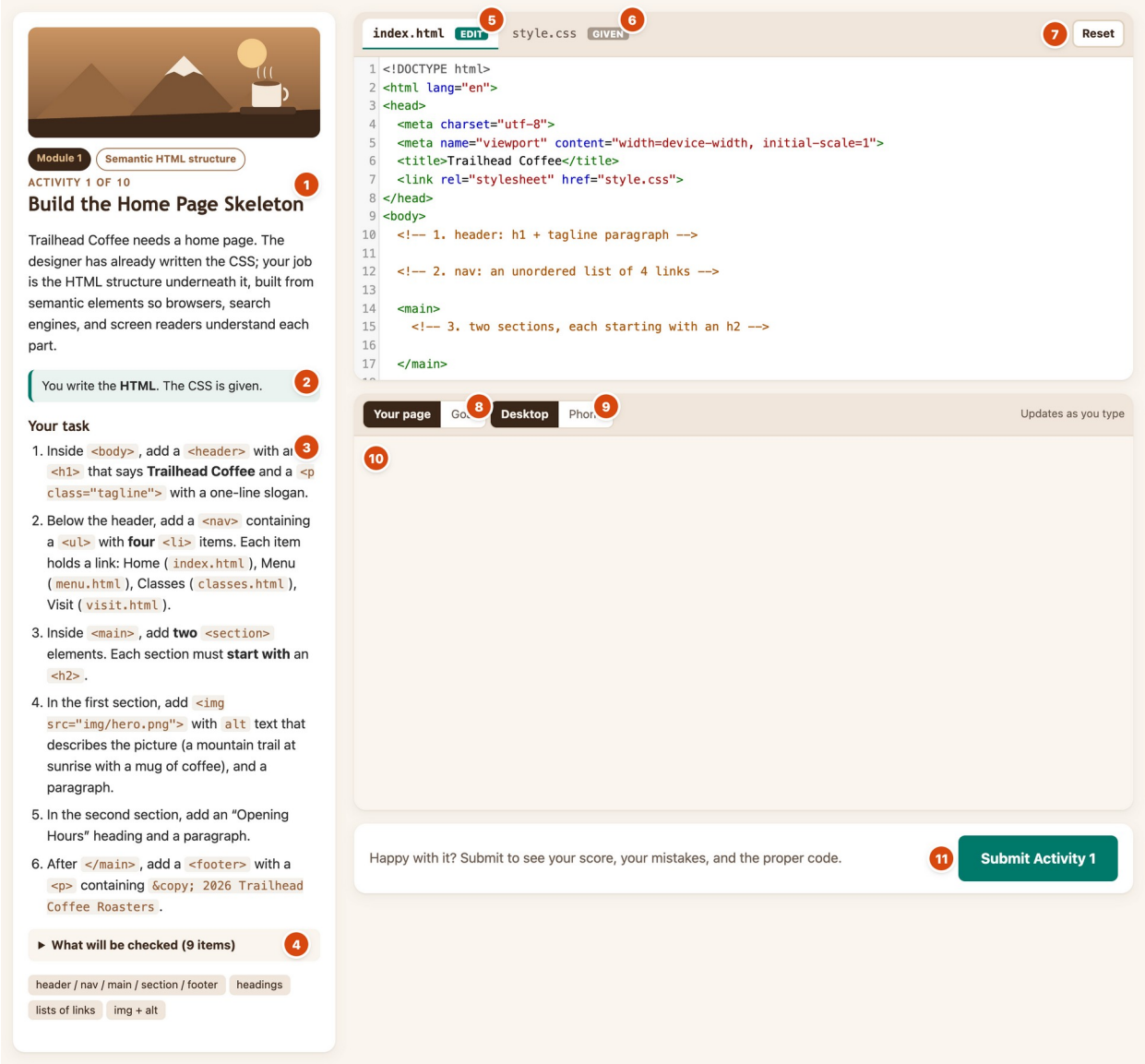


Figure 6. An activity screen. The numbers match the table below.

#	What it is	What to do with it
1	Title and story	The module, the topic, and a short Trailhead story explaining what the café needs.
2	Which file you write	Tells you whether you write the **HTML** or the **CSS** this time.
3	Your task	The numbered steps. Do them in order; each one names the tags, properties, or values to use.
4	What will be checked	Click to see every check and its points. Use it as a checklist before you submit.

#	What it is	What to do with it
5	The file you edit	Marked **EDIT**. Click in the code and type.
6	The given file	Marked **GIVEN**. You can read it, but not change it.
7	Reset	Puts the starter code back if you get lost. (It asks first.)
8	Your page / Goal	*Your page* shows your code; *Goal* shows what the finished page should look like.
9	Desktop / Phone	Shows the page at laptop width or at phone width (400px).
10	The preview	Updates about half a second after you stop typing.
11	Submit	Grades your code and shows your report.

The two files

A real web page has an HTML file and a CSS file, and so does every activity: index.html and style.css. In activities **1, 2, 9, and 10** you write the HTML; in activities **3 to 8** you write the CSS. Click the other tab to read the given file: it shows you the class names, IDs, and styles you are working with.



Figure 7. The given style.css in Activity 1. The yellow note reminds you that it is read-only.

Where is the <link> to the CSS?

The HTML already contains <link rel="stylesheet" href="style.css">. The lab connects the two files for you, exactly as a browser would.

Your page and the Goal

Click **Goal** at any time to see what you are aiming for. The goal shows only the finished **page**, not the code. Click **Your page** to go back.

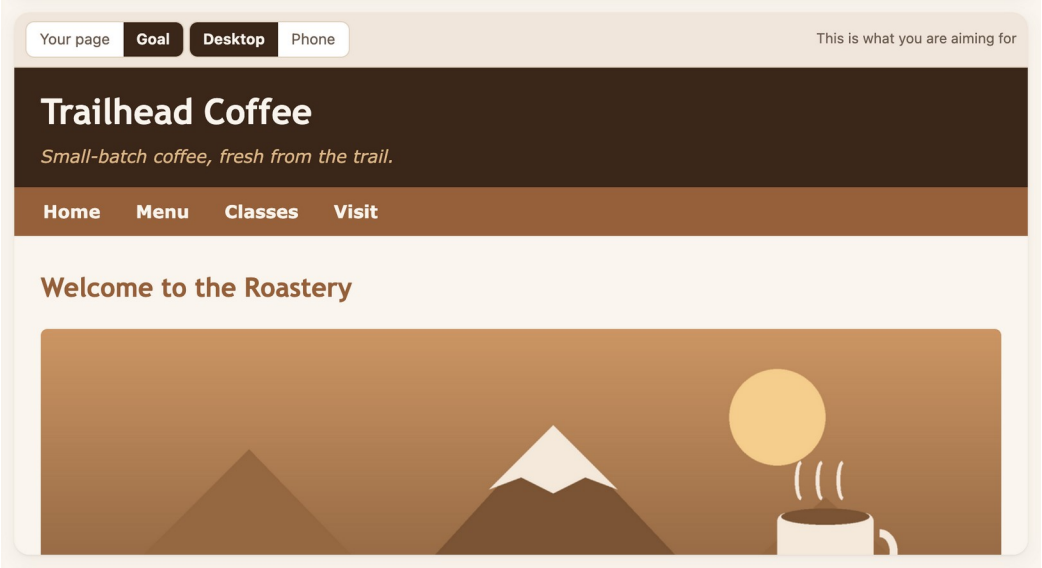


Figure 8. The Goal view of Activity 1.

Desktop and Phone

Click **Phone** to see your page at phone width. You need this most in **Activity 8**, where you make the layout change on small screens.

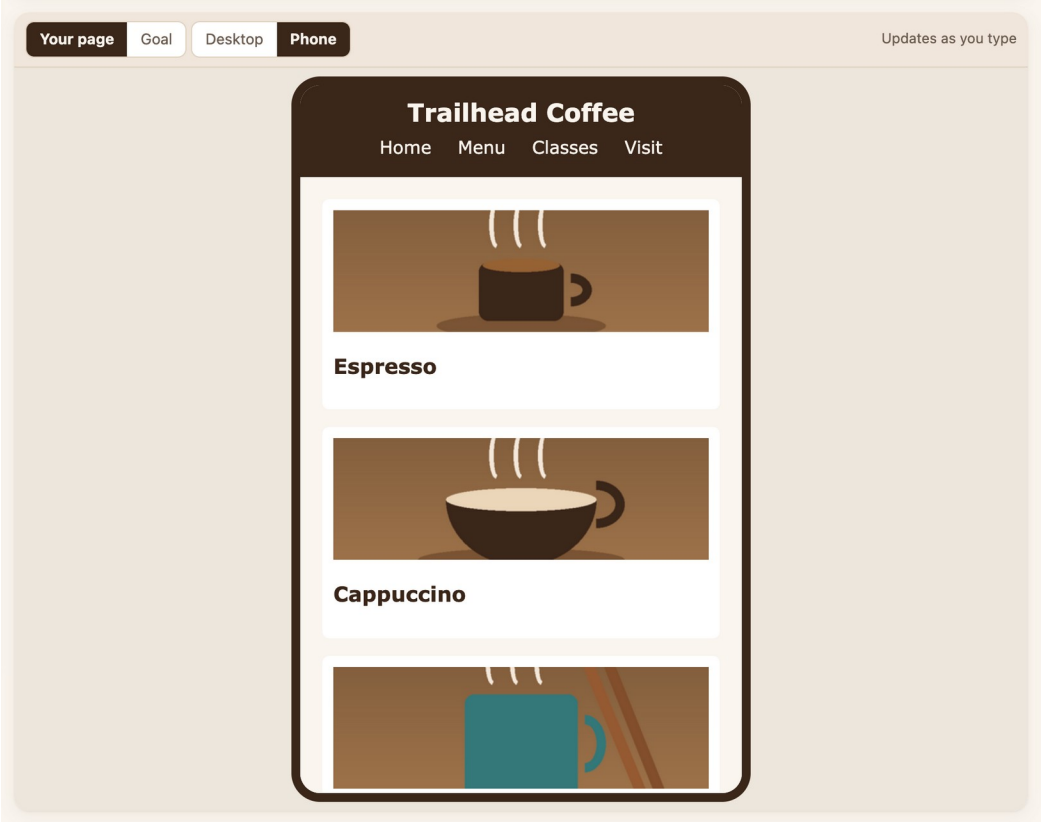


Figure 9. The Phone view in Activity 8, after adding a media query.

6. Your First Activity, Step by Step

Here is Activity 1 from start to finish, so you know exactly what to expect.

1. On the home page, type your name and click **Start Activity 1**.
2. Read the story and the six steps on the left. Open **What will be checked** to see the nine checks.
3. Click **Goal** to see the page you are building, then click **Your page** again.
4. In index.html, find the comment <!-- 1. header: h1 + tagline paragraph --> and write the header below it (you can delete the comment):

```
<header>
  <h1>Trailhead Coffee</h1>
  <p class="tagline">Small-batch coffee, fresh from the trail.</p>
</header>
```

5. Watch the preview: the brown header appears as soon as you stop typing.

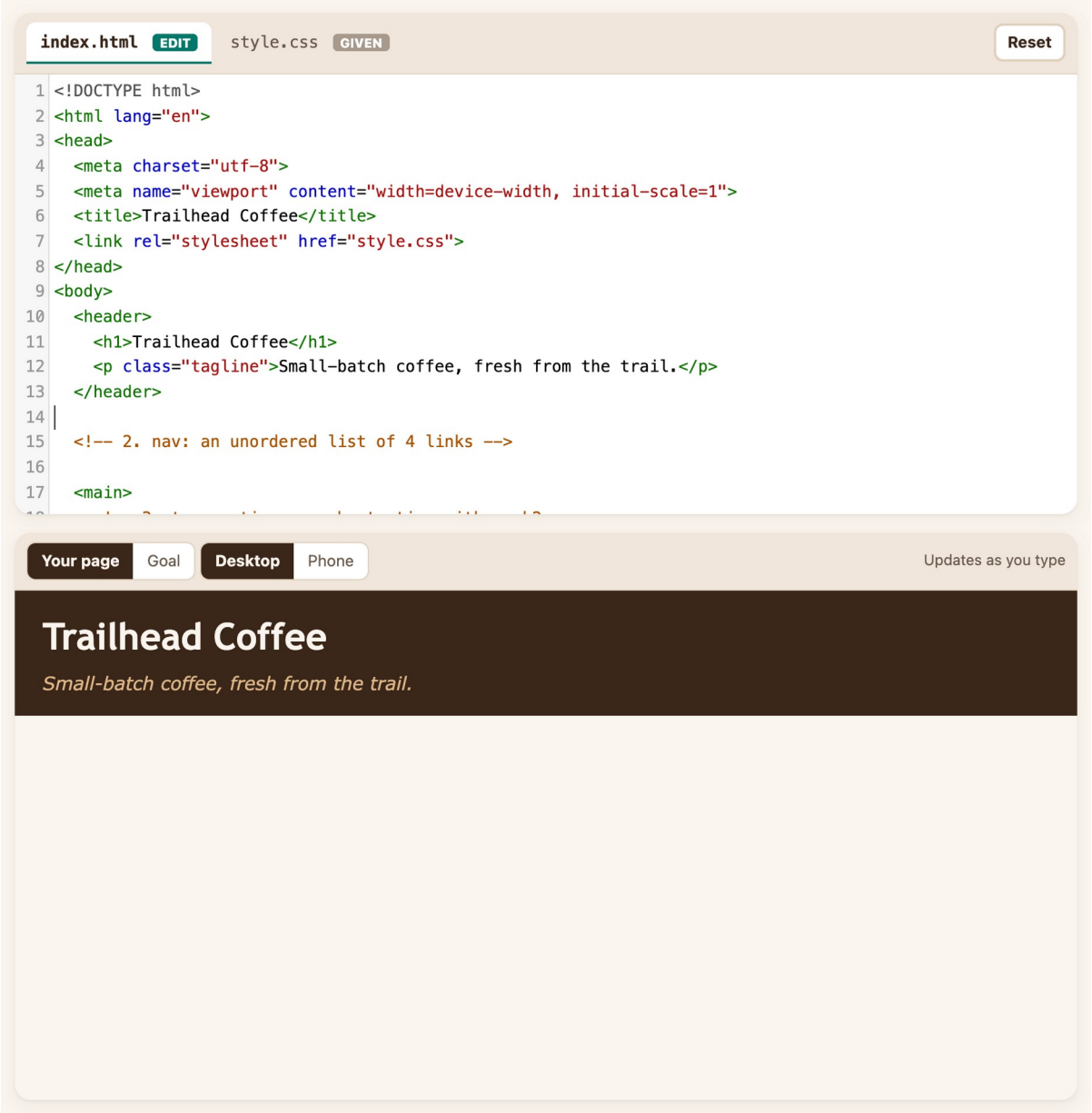


Figure 10. After step 4: the header appears in the preview. The given CSS already styles it.

6. Carry on with the nav, the two sections, the image, and the footer. Check the preview after each one.
7. When your page looks like the goal, click **Submit Activity 1**.
8. Read the report (Section 7). Fix anything that is wrong and click **Submit again**, or click **Next: Activity 2 →**.

Good habits

- Type the code; do not copy it. Typing is how the tags and properties stick.
- Keep your code tidy and indented. It is much easier to spot a missing closing tag.
- If the preview suddenly goes wrong, the last thing you typed is usually the problem: a missing `>`, `}`, or `;`.

7. Reading Your Report

After you submit, the report appears below your workspace. It is the most useful part of the lab, so read all of it.

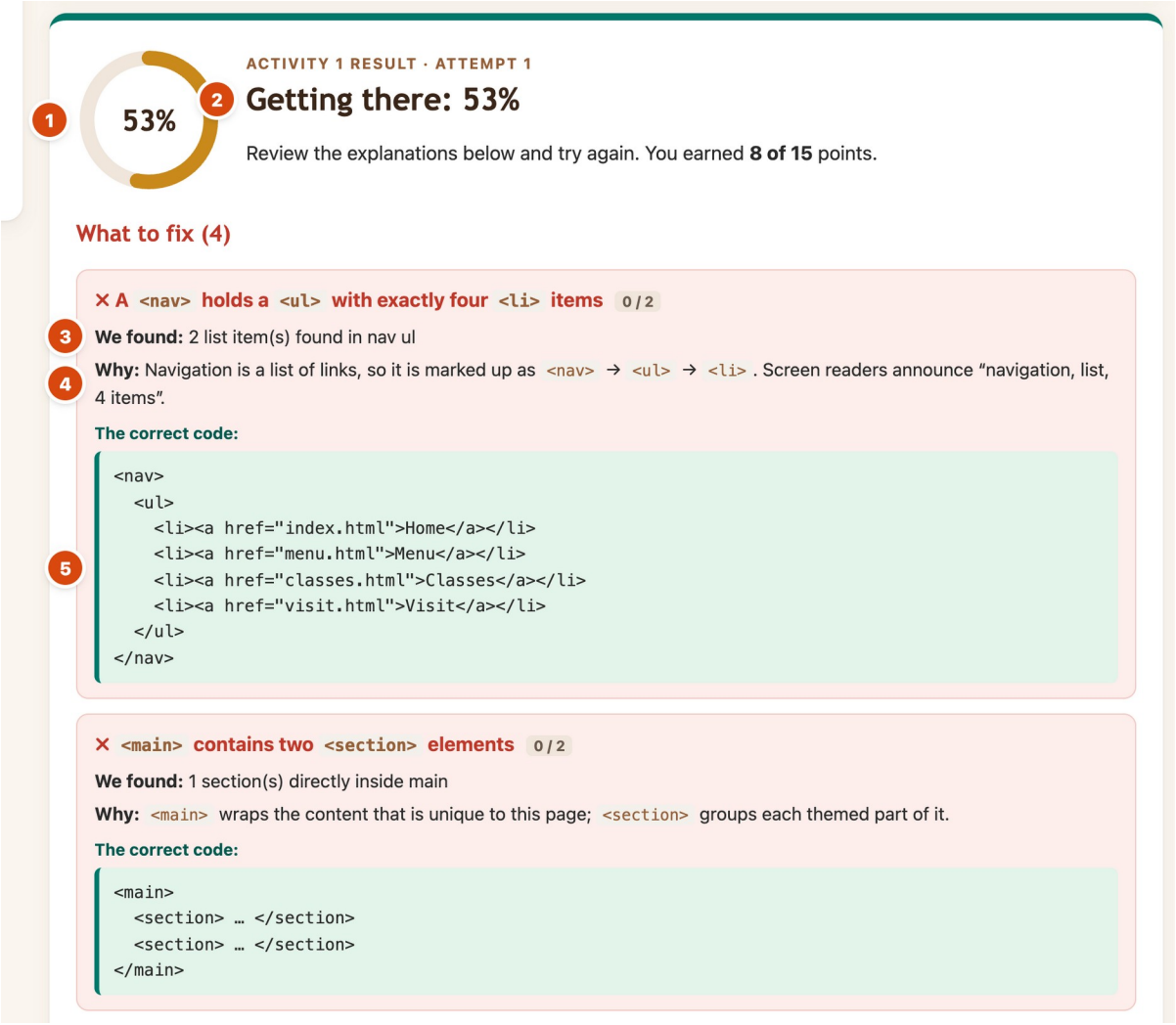


Figure 11. The top of a report. The numbers match the table below.

#	What it is	What to do with it
1	Your score	The percentage of points you earned. The colour shows how you did (see the table below).
2	Verdict	A one-line summary and your points, for example 8 of 15.
3	We found	What the lab actually found in your code, for example "2 list item(s) found in nav ul".
4	Why	Why this rule matters, in plain words.
5	The correct code	The code that passes this check. Compare it with yours, then fix your own code.

Score	Colour	Verdict	What to do
90–100%	Teal	Excellent	Move on.
75–89%	Green	Good work	Fix the small things, then move on.
50–74%	Amber	Getting there	Fix the mistakes and submit again.
0–49%	Red	Keep practising	Re-read the steps and the explanations, fix, and submit again.

Notes and the things you got right

Below the mistakes you may see a yellow box of **notes about your code**. These are not scored, but they point at problems such as a missing closing tag, an image with no `alt`, or a missing semicolon in CSS. Under that, **What you got right** lists the checks you passed, each with its explanation.

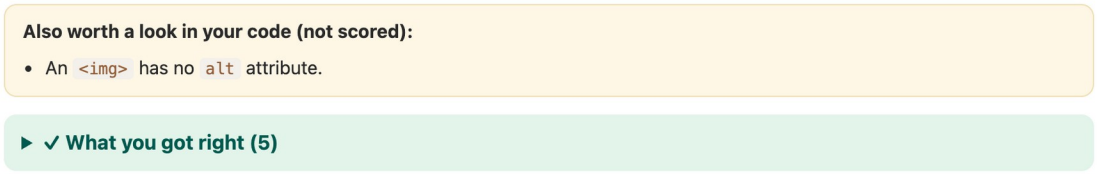


Figure 12. An unscored note, and the list of passed checks (click to open it).

Compare your code with the proper code

The report then shows your file and the proper file side by side, with matching lines lined up:

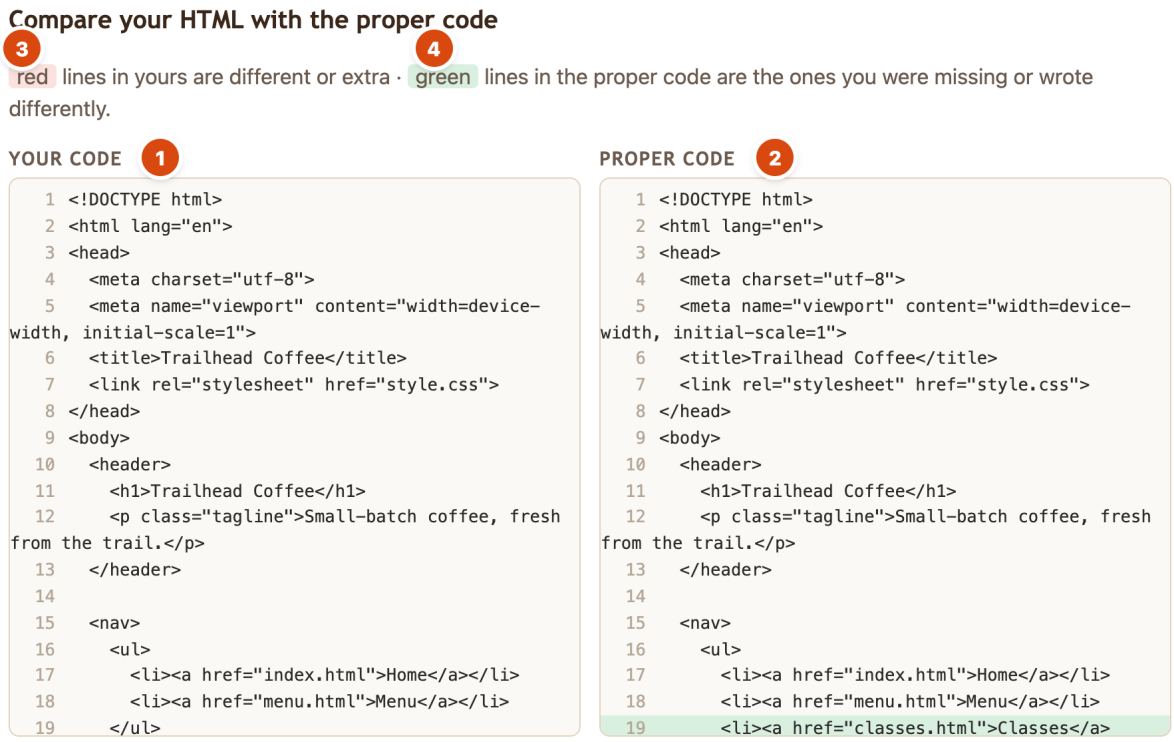


Figure 13. Your code (1) and the proper code (2). Red lines (3) are different or extra in yours; green lines (4) are what you were missing.

Finally, it shows **your page** next to the **goal**, and two buttons:

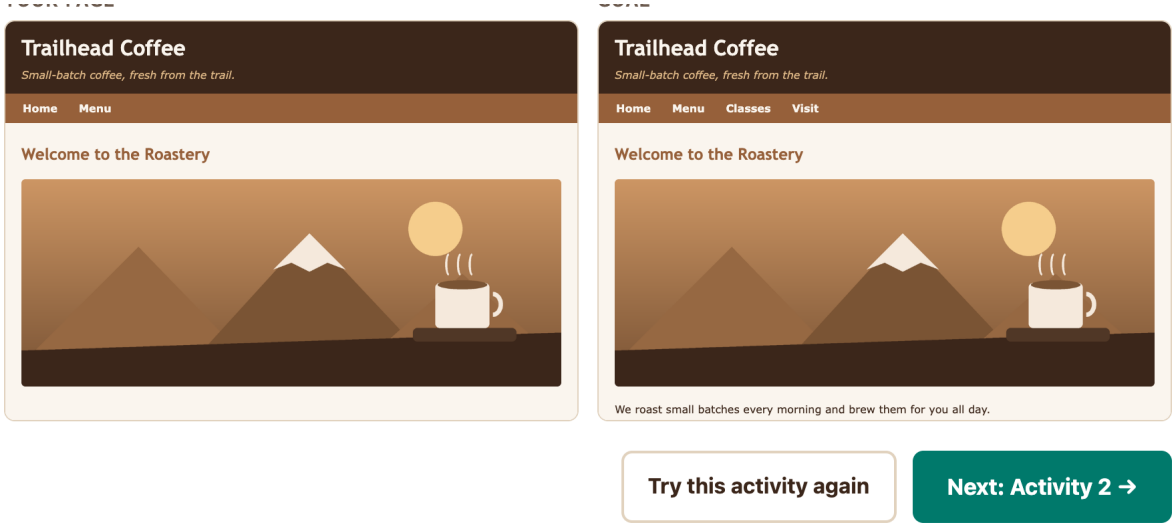


Figure 14. Your page and the goal, with the Try again and Next buttons.

- **Try this activity again** scrolls back to your code so you can fix it and submit again.
- **Next: Activity N →** goes on to the next activity, which is now unlocked.

8. Moving Through the Lab

As you submit activities, the circles at the top change colour, so you can see at a glance where you did well and what to revisit.



Figure 15. The activity circles after a full run: teal and green are strong; amber means "worth another try".

Circle	Meaning
Outlined, white border	The activity you are on now
Outlined, faded	Locked: submit the activity before it first
Teal / green	Submitted, 75% or more
Amber	Submitted, 50–74%
Red	Submitted, below 50%

You can click any open circle to go back to an earlier activity. Your last code is still there, and your last report is shown below it. Submit again to improve the score.

Two activities that work a little differently

- **Activity 8 (responsive design):** your CSS must change the layout **only** on small screens. Use the **Phone** button to test it, and make sure the **Desktop** view still looks like the goal.
- **Activity 10 (forms):** in the preview you can fill in your form and press its button. The preview then shows exactly what your form sent. A missing row usually means a control without a `name`.

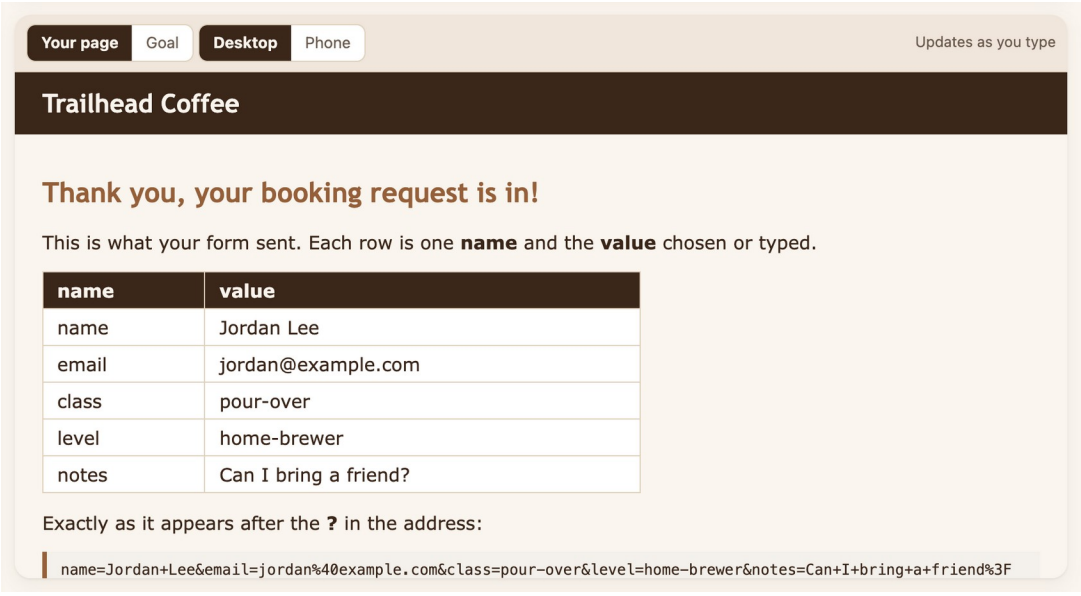


Figure 16. Activity 10: after pressing the form's button, the preview lists every name and value it sent.

9. Your Final Results

When you submit Activity 10, click **See my final results →** (or **My results** at the top) to see your full-cycle score.

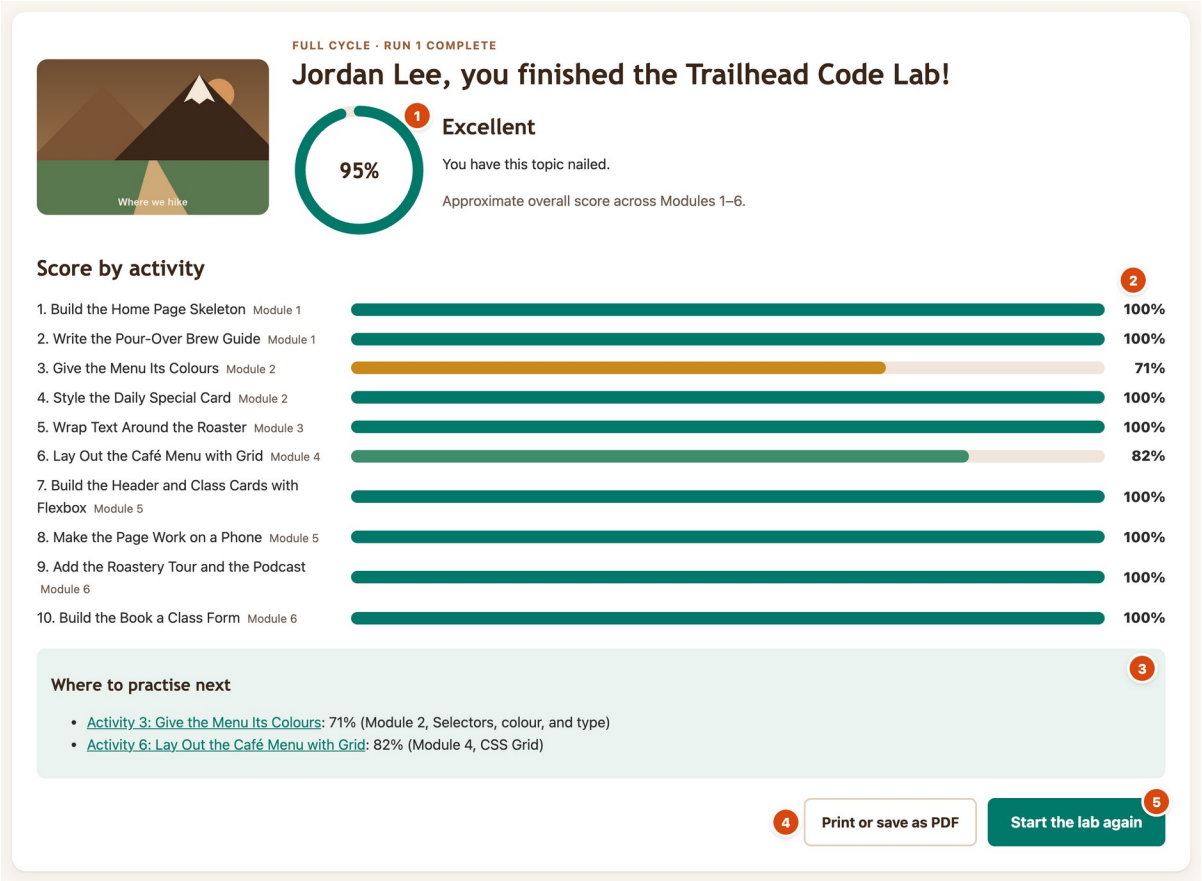


Figure 17. The results page after a full run. The numbers match the table below.

#	What it is	What to do with it
1	Overall score	The average of your ten activity scores, with a verdict.
2	Score by activity	One bar per activity. Click a bar to go back to that activity.
3	Where to practise next	Your lowest-scoring activities, with links.
4	Print or save as PDF	Opens your browser's print window. Choose **Save as PDF** to keep a copy or hand it in.
5	Start the lab again	Saves this run in your history and starts a fresh run from Activity 1.

Doing the lab again

A new run starts every activity from its starter code, so it is a real second practice, ideal before a quiz. Your earlier runs and your best overall score are listed on the home page under **Your earlier runs**.

10. Tips for Success

1. **Read all the steps before you type.** Many mistakes come from missing a word such as “inside”, “first”, or “only”.
2. **Look at the given file.** In CSS activities the HTML tells you the class names; in HTML activities the CSS tells you which classes it expects (for example `.tagline`).
3. **Use the checklist.** Open *What will be checked* before you submit and tick each item off in your head.
4. **Check both previews.** Compare *Your page* with *Goal*, and in Activities 7 and 8 try *Phone* too.
5. **Spelling counts.** `class="tagline"` and `class="Tagline"` are different; so are `img/hero.png` and `images/hero.png`.
6. **One mistake can hide others.** A missing `}` or `;` can switch off every rule after it. The yellow notes in the report point these out.
7. **Do not rush to the proper code.** Try to fix each mistake from the *Why* first; then compare.
8. **Short sessions work well.** Two or three activities at a time is plenty. Your progress waits for you.

11. Troubleshooting

Problem	What to do
My progress has disappeared.	You are probably in a different browser, on a different computer, or in a private window. Go back to the one you used before.
An activity says it is locked.	Submit the activity before it. Each one unlocks the next.
The preview is blank or looks broken.	Look at the last thing you typed. A missing `>` or an unclosed tag in HTML, or a missing `}` in CSS, can break everything after it.
I cannot type in the code.	You are on the **GIVEN** tab. Click the tab marked **EDIT**.
My page looks exactly like the goal but I lost points.	Read *We found*. Some checks look at the code itself: for example `alt` text, `name` attributes, or using `em` instead of `px`.
The video in Activity 9 does not play in the preview.	That is fine: the checks look at your HTML, not at the video playing. Check your `src` paths against the step list.
I want to start an activity over.	Click **Reset** above the code.
I want to start the whole lab over.	Finish the run, then click **Start the lab again** on the results page.

Still stuck? Bring a screenshot of your code and your report to class or office hours.

Quick Reference

Button or label	What it does
Start / Continue (home page)	Opens the next activity you have not submitted
EDIT / GIVEN tabs	The file you write / the file you can only read
Your page / Goal	Your result / what you are aiming for
Desktop / Phone	Preview at laptop width / at phone width
Reset	Puts the starter code back
Submit Activity N	Grades your code and shows your report
Try this activity again	Back to your code to fix it and resubmit
Next: Activity N →	Opens the next activity
My results	Scores so far, or the full-cycle results
Print or save as PDF	Saves your results page
Start the lab again	Saves this run in your history and starts a new run

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